Seed geometry in nucleation-controlled transition times in time-dependent Ginzburg–Landau models

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Abstract

Generalized Mpemba effects concern anomalous relaxation in which a system prepared farther from equilibrium can reach a target condition faster than a system prepared closer to equilibrium. Phase transitions are a stringent setting for this idea because the relevant time scale can be controlled not only by smooth relaxation but also by nucleation of a new phase. Motivated by this problem, we study transition times in spatially extended time-dependent Ginzburg–Landau (TDGL) models. We first use a continuous-transition ϕ^4 model as a baseline and then analyze a first-order ϕ^6 model in which the post-quench state is metastable. In the first-order model, larger initial-state labels generate more barrier-crossing seed candidates, as measured by the fraction and largest cluster size of sites satisfying $|\phi| > \phi_b$. However, these seed candidates become less compact and more spatially extended as the initial-state label increases. Ordered-like initial seeds close to the stable phase are almost absent, and the global transition time is closely associated with the stochastic cluster-nucleation time once both are observed. The results show that the amount of barrier-crossing field is not by itself sufficient to characterize nucleation-controlled transition times. Together, they establish seed geometry as an independent diagnostic, distinct from seed amount, for interpreting nucleation-controlled transition times. The work identifies a concrete mechanism by which increasing the amount of barrier-crossing field can fail to produce a faster first-order transition: the additional seed material may be spatially less organized as a nucleation precursor.

Keywords: Mpemba effect; time-dependent Ginzburg–Landau equation; first-order phase transition; nucleation; seed compactness; nonequilibrium relaxation

I. INTRODUCTION

The Mpemba effect refers to the counterintuitive observation that, under certain conditions, a system prepared at a higher initial temperature can reach a colder final state faster than an otherwise identical system prepared at a lower initial temperature [1]. Although the phenomenon is historically associated with the freezing of water, its reproducibility and microscopic origin in water remain subtle because evaporation, convection, dissolved gases, impurities, supercooling, and nucleation may all affect the freezing time [2–4]. For this reason, recent studies increasingly treat the Mpemba effect as a more general nonequilibrium relaxation phenomenon rather than as a peculiarity of water alone [5].

In modern terms, a generalized Mpemba effect occurs when a system prepared further from equilibrium relaxes faster than a system prepared closer to equilibrium after both are quenched to the same final condition [6, 7]. Controlled experiments on colloidal systems have demonstrated such anomalous relaxation in settings where the relevant state variables and thermal bath are well characterized [8]. These developments have motivated theoretical studies that relate the effect to the spectral structure of relaxation dynamics, the overlap of the initial state with slow modes, and the geometry of relaxation pathways in state space.

A particularly important case is the Mpemba effect through a phase transition. In many experimentally motivated situations, including freezing, the relevant observable is not merely the relaxation of an average temperature but the time at which a new phase appears or becomes dominant. Holtzman and Raz formulated this idea using Landau theory and introduced a phase-transition time as a concrete observable for identifying Mpemba-like behavior through a continuous transition [9]. Related mean-field studies have also considered phase-transition times as the relevant observable for Mpemba-like behavior [10]. First-order transitions provide another natural setting because their kinetics are typically controlled by metastability, interfacial free-energy costs, and nucleation [11–15] rather than by deterministic linear growth alone. Zhang and Hou proposed a theoretical model for Mpemba-like behavior through a canonical first-order phase transition, showing that different initial temperatures may follow different transition pathways and thereby exhibit inverted transition times [16].

Despite these advances, the role of spatial structure in first-order, nucleation-controlled phase-transition protocols remains incompletely understood. In spatially extended systems, the existence of local fluctuations beyond a free-energy barrier does not automatically imply the formation of a growing nucleus. A local seed must also have sufficient size, compactness, and proximity to the stable phase to overcome the interfacial cost associated with forming a domain boundary [15, 17, 18]. This distinction is especially relevant for first-order transitions, where subcritical droplets may frequently appear and disappear without triggering a macroscopic transition. Related studies in spin systems have also emphasized that spatial correlations in the initial state can strongly influence the apparent Mpemba effect [19, 20].

In this work, we investigate this issue using time-dependent Ginzburg–Landau dynamics in spatially extended scalar field models. We compare two minimal settings. The first is a continuous-transition ϕ^4 model, used as a baseline in which ordering is driven by the growth of unstable long-wavelength modes. The second is a first-order ϕ^6 model, in which the post-quench state is metastable and the transition requires nucleation of the stable phase.

The central question of this work is how a temperature-like initial-state label affects the transition time when the transition is controlled by nucleation. More specifically, we ask whether seed-amount observables are sufficient to characterize the observed finite-time nucleation probability, or whether spatial information about the seed is also needed.

Our numerical results show that the answer is nontrivial. In the first-order model, larger initial-state labels generate more barrier-crossing seeds, as measured by the fraction and maximum cluster size of sites satisfying $|\phi| > \phi_b$, where ϕ_b is the barrier location. However, these seeds are not necessarily ordered-phase droplets. Ordered-like initial seeds, defined by $|\phi| > 0.5\phi_s$, are almost absent. Moreover, the largest barrier-crossing seed becomes less compact and more spatially extended as the initial-state label increases. Consequently, the observed finite-time nucleation probability is not captured by the monotonic increase in the number of barrier-crossing seeds.

The purpose of the present paper is therefore not to claim that the larger-label initial state always transforms faster. Rather, we use a minimal TDGL model to show that transition-time behavior in a nucleation-controlled setting cannot be understood from seed amount alone and requires spatial information about seed quality. In particular, seed compactness and post-quench stochastic nucleation can modify or obscure a simple monotonic relation between the initial-state label and transition time.

The remainder of this paper is organized as follows. In Sec. II, we define the TDGL models, the quench protocol, and the observables used to detect transition and nucleation times. In Sec. III, we present the numerical results, starting from the continuous-transition baseline and then analyzing nucleation, seed amount, and seed quality in the first-order model. In Sec. IV, we discuss the physical interpretation, limitations, and relation to phase-transition Mpemba effects. Section V summarizes the conclusions.

II. MODEL AND NUMERICAL PROTOCOL

A. TDGL dynamics

We consider a two-dimensional scalar order parameter field,

$$\phi = \phi(\mathbf{x}, t), \tag{1}$$

evolving under overdamped time-dependent Ginzburg–Landau dynamics,

$$\frac{\partial \phi}{\partial t} = \nabla^2 \phi - \frac{\partial V(\phi)}{\partial \phi} + \eta(\mathbf{x}, t). \tag{2}$$

This is a Model-A-type relaxational dynamics for a nonconserved scalar order parameter [18, 21, 22].

Here, $V(\phi)$ is a local Landau free-energy density and η is a Gaussian white noise term. Unless otherwise stated, the noise satisfies

$$\langle \eta(\mathbf{x}, t) \rangle = 0, \tag{3}$$

$$\langle \eta(\mathbf{x}, t) \eta(\mathbf{x}', t') \rangle = 2D\delta(\mathbf{x} - \mathbf{x}')\delta(t - t'). \tag{4}$$

The coefficient D controls the strength of stochastic fluctuations. In the present work, D_i denotes the noise strength used to prepare the initial state, while D_f denotes the post-quench noise strength.

The spatial Laplacian term corresponds to a gradient-energy contribution,

$$F[\phi] = \int d^2x \left[\frac{1}{2} |\nabla \phi|^2 + V(\phi) \right]. \tag{5}$$

This term penalizes sharp spatial variations in ϕ. It therefore generates an interfacial cost between regions in different phases, which is crucial for nucleation.

B. Quench protocol

The simulations use the following protocol. First, an initial field is prepared by evolving the TDGL equation under an initial potential $V_i(\phi)$ and an initial noise strength D_i. Second, at $t = 0$, the initial parameters are suddenly replaced by final parameters $V_f(\phi)$ and D_f. Third, the subsequent relaxation, nucleation, and transition of the field are monitored.

This sudden parameter change is referred to as a quench. Physically, it represents an idealized rapid cooling or rapid change of thermodynamic control parameters. Initial states are labeled by a temperature-like parameter

$$T_i \in \{1.05, 1.10, 1.20, 1.50, 2.00, 3.00\}. \tag{6}$$

In the current implementation, T_i should be interpreted as an initial-state label rather than as a calibrated physical temperature. In particular, we do not attempt to impose a unique thermodynamic temperature scale for the nonequilibrium preparation dynamics; T_i rescales the initial fluctuation strength used to generate the pre-quench ensemble. For the first-order model, the main initialization scheme fixes the initial mass parameter a_i and changes the initial fluctuation strength as

$$D_i = D_0 T_i. \tag{7}$$

This isolates the effect of stronger initial fluctuations while keeping the pre-quench potential structure fixed.

C. Continuous-transition baseline: ϕ^4 model

As a baseline, we consider a continuous-transition TDGL model with

$$V_4(\phi; r) = \frac{r}{2}\phi^2 + \frac{1}{4}\phi^4. \tag{8}$$

The initial state is prepared with $r_i > 0$, for which $\phi = 0$ is stable. After the quench, the final potential is

$$V_{4,f}(\phi) = -\frac{1}{2}\phi^2 + \frac{1}{4}\phi^4. \tag{9}$$

The post-quench dynamics is therefore

$$\frac{\partial \phi}{\partial t} = \nabla^2 \phi + \phi - \phi^3 + \eta_f. \tag{10}$$

At early times and small ϕ, this reduces to the linearized equation

$$\frac{\partial \phi_{\mathbf{k}}}{\partial t} = (1 - k^2)\phi_{\mathbf{k}}, \tag{11}$$

so that modes with $k < 1$ grow exponentially. Thus, in this model, the early ordering dynamics is controlled by the initial overlap with unstable long-wavelength modes.

We monitor the global ordering measure

$$Q(t) = \langle \phi^2(\mathbf{x}, t) \rangle, \tag{12}$$

where $\langle \cdots \rangle$ denotes the spatial average. The transition time is defined operationally as the first time at which

$$Q(t) > Q_{\rm th}. \tag{13}$$

For the baseline calculations, we use $Q_{\rm th} = 0.5$.

D. First-order transition model: ϕ^6 potential

To study nucleation-controlled transition times, we use a first-order Landau potential,

$$V_6(\phi) = \frac{a}{2}\phi^2 - \frac{b}{4}\phi^4 + \frac{c}{6}\phi^6, \tag{14}$$

with $b > 0$ and $c > 0$. In most simulations we set $b = c = 1$, so that a is the main control parameter.

The extrema satisfy

$$\frac{\partial V_6}{\partial \phi} = a\phi - b\phi^3 + c\phi^5 = 0. \tag{15}$$

Besides $\phi = 0$, the nonzero extrema satisfy

$$c\phi^4 - b\phi^2 + a = 0. \tag{16}$$

When $0 < a < b^2/(4c)$, the two nonzero extrema are given by

$$\phi_b^2 = \frac{b - \sqrt{b^2 - 4ac}}{2c}, \tag{17}$$

$$\phi_s^2 = \frac{b + \sqrt{b^2 - 4ac}}{2c}. \tag{18}$$

Here, ϕ_b is the barrier location and ϕ_s is the stable-phase amplitude. For $0 < a < 3b^2/(16c)$, the state $\phi = 0$ is metastable and $\phi = \pm\phi_s$ are the globally stable phases. This is the regime used for the first-order post-quench dynamics.

The post-quench equation is

$$\frac{\partial \phi}{\partial t} = \nabla^2\phi - a_f\phi + b\phi^3 - c\phi^5 + \eta_f. \tag{19}$$

In the main simulations, we use

$$a_f = 0.02, \quad b = c = 1. \tag{20}$$

E. Seed definitions

For the first-order model, we distinguish several notions of seed.

A barrier-crossing seed is defined as an initial region satisfying

$$|\phi(\mathbf{x},0)| > \phi_b. \tag{21}$$

This identifies local fluctuations that have crossed the free-energy barrier in field amplitude. We define

$$p_{\text{seed}} = \frac{1}{N^2}\sum_{\mathbf{x}} \mathbf{1}\left(|\phi(\mathbf{x},0)| > \phi_b\right), \tag{22}$$

and

$$c_{\text{seed}} = \max_{\text{clusters}} \left[\text{cluster size of } |\phi(\mathbf{x},0)| > \phi_b\right]. \tag{23}$$

These observables measure the amount and size of barrier-crossing seed candidates. However, they do not by themselves imply the existence of a growing nucleus.

To detect initial regions already close to the stable phase, we also define ordered-like seeds by

$$|\phi(\mathbf{x},0)| > 0.5\phi_s. \tag{24}$$

The corresponding observables are $p_{\text{ordered seed}}$ and $c_{\text{ordered seed}}$, defined analogously to p_{seed} and c_{seed}. The threshold $0.5\phi_s$ is an operational choice. It is stricter than the barrier threshold ϕ_b, but less restrictive than requiring the field to reach ϕ_s itself.

For the largest barrier-crossing seed cluster, we compute a compactness measure,

$$C_{\text{comp}} = \frac{A_{\text{seed}}}{A_{\text{bbox}}}, \tag{25}$$

where A_{seed} is the cluster area and A_{bbox} is the area of the smallest unwrapped bounding box containing that cluster. A compact cluster has C_{comp} closer to unity, while a string-like or spatially extended cluster has a smaller value. We also compute the radius of gyration,

$$R_g^2 = \frac{1}{A_{\text{seed}}}\sum_{\mathbf{x}\in\text{seed}} |\mathbf{x}-\mathbf{x}_{\text{cm}}|^2, \tag{26}$$

where $\mathbf{x}_{\text{cm}}$ is the cluster center of mass in the unwrapped coordinates. Together, compactness and R_g quantify the spatial quality of a seed.

F. Nucleation and transition times

The nucleation time is defined using clusters of ordered-like regions during the post-quench dynamics. At each measurement time, we identify sites satisfying

$$|\phi(\mathbf{x}, t)| > 0.5\phi_s. \tag{27}$$

Let $C_{\max}(t)$ be the maximum cluster size of such ordered-like sites. We define the cluster nucleation time t_{nuc} as the first time at which

$$C_{\max}(t) \geq C_{\text{th}} \tag{28}$$

for at least n_{cons} consecutive measurements. In the main calculations,

$$C_{\text{th}} = 20, \quad n_{\text{cons}} = 3. \tag{29}$$

This persistence requirement reduces false positives from short-lived stochastic fluctuations. Throughout the paper, P_{nuc} denotes the empirical probability that this criterion is satisfied within the finite observation window, not an extrapolated infinite-time nucleation probability.

The transition time is defined from the global order measure $Q(t) = \langle \phi^2(\mathbf{x}, t) \rangle$. For the first-order model, we use

$$Q_{\text{th}} = q_{\text{frac}}\phi_s^2, \quad q_{\text{frac}} = 0.5. \tag{30}$$

Thus,

$$t_{\text{tr}} = \min\{t : Q(t) > Q_{\text{th}}\}. \tag{31}$$

The nucleation time t_{nuc} captures the first formation of a persistent ordered cluster, while t_{tr} measures the later, more global progress of the transition.

G. Numerical discretization

The TDGL equation is solved on a two-dimensional periodic grid of size $N \times N$. Most main calculations use $N = 64$ and $\Delta x = 1$. The deterministic linear part is treated semi-implicitly in Fourier space, while the nonlinear terms and noise are treated explicitly. For an equation of the form

$$\frac{\partial \phi}{\partial t} = \nabla^2 \phi + \alpha\phi + \mathcal{N}(\phi) + \eta, \tag{32}$$

the Fourier-space update has the schematic form

$$\phi_{\mathbf{k}}(t+\Delta t) = \frac{\phi_{\mathbf{k}}(t) + \Delta t\, \mathcal{N}_{\mathbf{k}}(t) + \Delta W_{\mathbf{k}}}{1 + \Delta t(k^2 - \alpha)}. \tag{33}$$

Here $\Delta W_{\mathbf{k}}$ is the Fourier transform of the real-space stochastic increment. Equation (33) is meant as a schematic representation of the semi-implicit update; in the code the Fourier wave numbers are those appropriate to the periodic grid and the chosen discretization of the Laplacian. In the discrete implementation, the real-space stochastic increment at each time step is generated with variance proportional to $2D\Delta t/\Delta x^2$. The timestep used in the main first-order runs is $\Delta t = 0.02$. Robustness checks with a smaller timestep, larger system size, longer pre-equilibration, and alternative nucleation thresholds are reported in the Supplemental Material.

III. RESULTS

In this section, we first examine the continuous-transition ϕ^4 TDGL model as a baseline. We then analyze the nucleation probability, the amount of barrier-crossing seeds, and the spatial quality of seeds in the first-order ϕ^6 TDGL model.

The central result is that larger initial-state labels contain more barrier-crossing seed candidates, but those candidates become less compact and more spatially extended. Therefore, the observed finite-time nucleation probability cannot be characterized by seed-amount observables alone.

A. Continuous-transition baseline

We first consider the continuous-transition ϕ^4 TDGL model as a baseline. After the quench, the state $\phi = 0$ becomes linearly unstable, and the early ordering dynamics is controlled by the overlap of the initial field with unstable growing modes.

Figure 1 shows the mean transition time t_{tr} as a function of the initial-state label. Under the present standard initialization, t_{tr} increases with the initial-state label. Thus, a simple Mpemba-like inversion, in which the larger-label initial state transforms faster, is not observed in the continuous-transition ϕ^4 baseline.

This result provides a useful reference point for the first-order model discussed below. In the standard continuous-transition TDGL model, increasing the initial-state label does

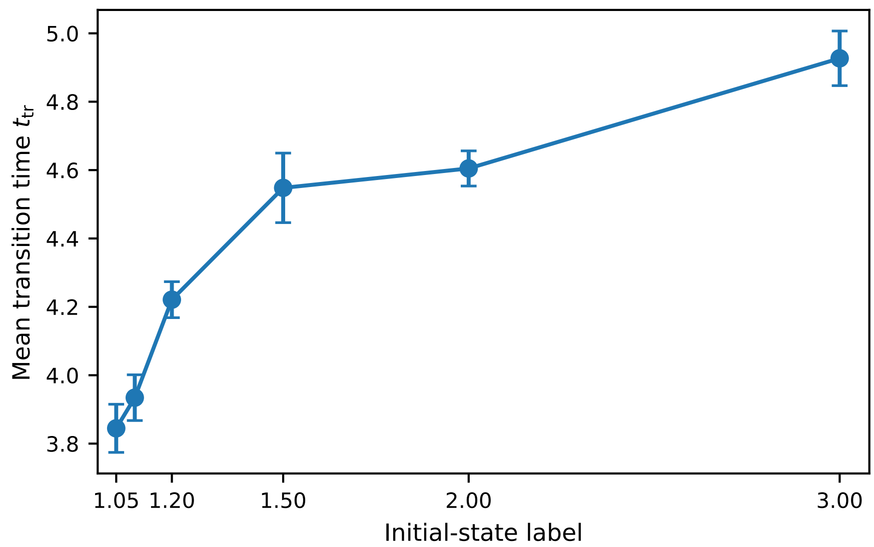

FIG. 1. Mean transition time in the continuous-transition ϕ^4 baseline model. Error bars indicate the standard error over samples. Under the present initialization, the transition time increases with the initial-state label, and a simple Mpemba-like inversion is not observed.

not automatically accelerate the transition. Therefore, the behavior of the first-order ϕ^6 model must be interpreted in terms of nucleation rather than simple deterministic growth of unstable modes.

B. Nucleation-controlled transition in the first-order model

We next study the first-order ϕ^6 TDGL model. In the post-quench potential, $\phi = 0$ is metastable and $\phi = \pm\phi_s$ are stable phases. The transition therefore proceeds through nucleation out of the metastable state rather than through a linear instability.

Figure 2(a) shows the empirical finite-time nucleation probability P_{nuc} as a function of the initial-state label. Here P_{nuc} is the fraction of samples in which the operational nucleation criterion is met by the final observation time. It does not increase monotonically with the initial-state label. This is important because, as shown below, the amount of barrier-crossing seeds increases monotonically with the initial-state label. Thus, the observed nucleation probability is not captured by the amount of such seeds alone.

Figure 2(b) shows the survival curves of the nucleation time t_{nuc}. Samples that do not nucleate by the final observation time are treated as right-censored. The survival curves are

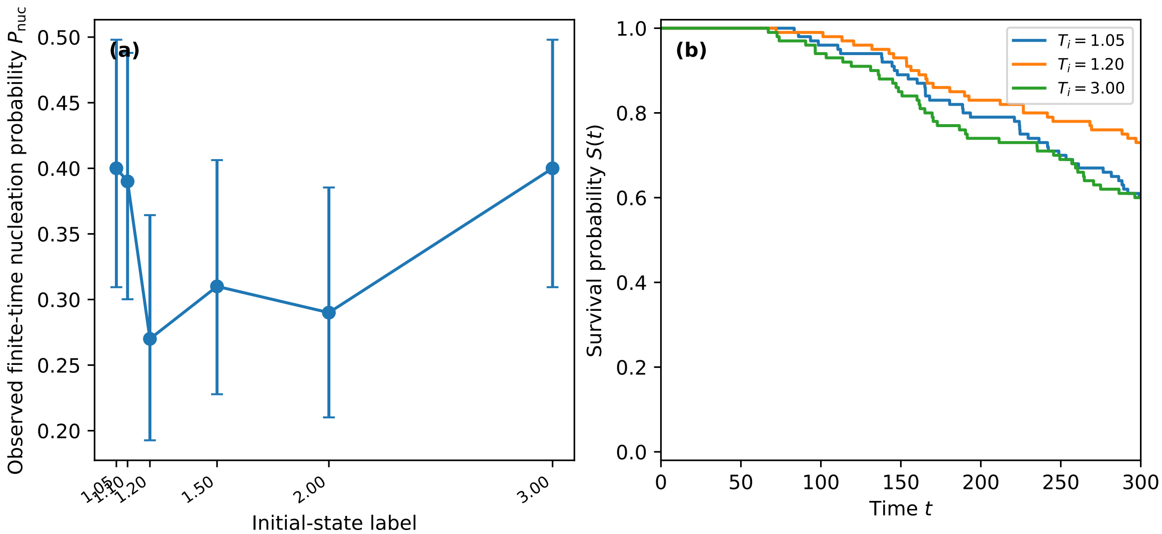

FIG. 2. Nucleation-controlled transition in the first-order ϕ^6 model. (a) Observed finite-time nucleation probability as a function of the initial-state label. Error bars indicate 95% Wilson binomial confidence intervals. (b) Right-censored survival curves of the nucleation time. Samples that do not nucleate by the final observation time are treated as censored.

product-limit curves with right-censoring at the final observation time [23]. They show that nucleation is broadly distributed in time and strongly stochastic. The initial-state label alone does not uniquely determine the nucleation time, and samples without a detected event by the final time remain censored rather than being assigned an artificial transition time.

These results are consistent with the transition time in the first-order ϕ^6 model being controlled by a combination of the initial seed structure and stochastic post-quench nucleation dynamics. A direct comparison between $t_{\rm nuc}$ and $t_{\rm tr}$ for samples in which both times were observed is shown below in Fig. 5(b). The strong correlation between these two times, quantified below, supports the interpretation that the global transition time is closely tied to the stochastic nucleation time in the first-order model.

C. Barrier-crossing seed amount

In a first-order transition, local fluctuations beyond the free-energy barrier can act as seed candidates. We therefore define barrier-crossing seeds by

$$|\phi(\mathbf{x},0)| > \phi_b, \tag{34}$$

where ϕ_b is the barrier location.

Figure 3(a) shows the barrier-crossing seed fraction p_{seed}. This quantity increases monotonically with the initial-state label. Thus, larger initial-state labels contain more local fluctuations that cross the free-energy barrier.

Figure 3(b) shows the largest barrier-crossing cluster size c_{seed}. This quantity also increases monotonically with the initial-state label. Therefore, larger initial-state labels not only contain more barrier-crossing sites, but also contain larger connected barrier-crossing clusters.

However, comparing Fig. 2(a) and Fig. 3 reveals an important point. While both p_{seed} and c_{seed} increase monotonically, P_{nuc} does not show the same clear monotonic trend within the present statistics. This shows that seed-amount observables alone are insufficient to explain the observed finite-time nucleation probability.

D. Seed quality and spatial structure

A barrier-crossing seed is not necessarily a growing nucleus. In a first-order transition, a stable-phase domain must overcome the interfacial cost associated with forming a boundary with the surrounding metastable phase, as emphasized in classical nucleation theory [11–15]. Therefore, not only the amount but also the spatial quality of the seed is important.

Figure 4(a) shows the compactness of the largest barrier-crossing seed cluster. The compactness decreases with the initial-state label. Thus, although larger initial-state labels contain more and larger barrier-crossing seeds, those seeds become less compact.

Figure 4(b) shows the radius of gyration R_g of the largest barrier-crossing seed cluster. R_g increases with the initial-state label, showing that seeds at larger initial-state labels are more spatially extended.

Together, Fig. 3 and Fig. 4 indicate the following picture:

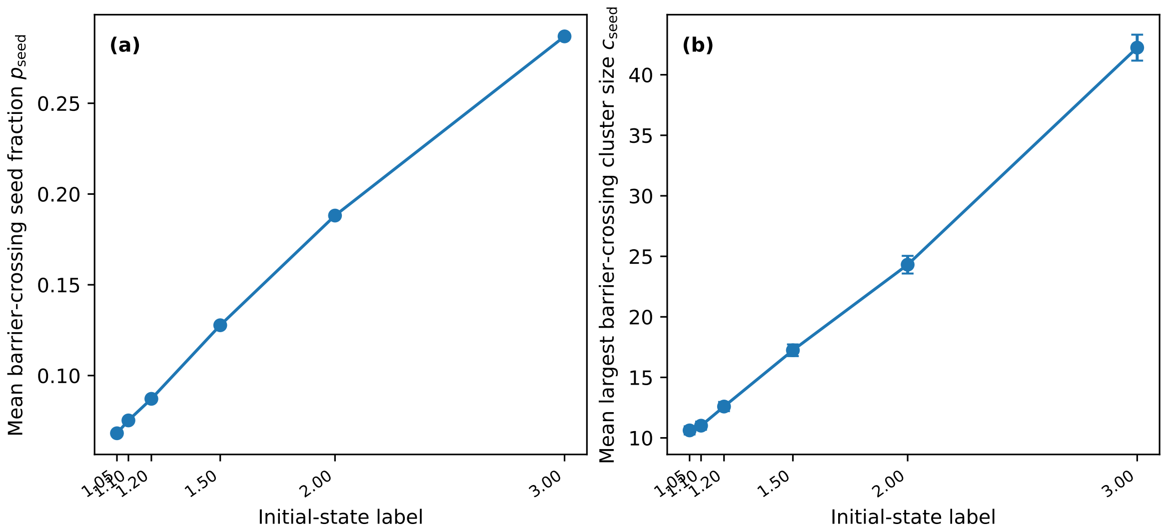

FIG. 3. Amount of barrier-crossing seeds in the first-order ϕ^6 model. (a) Mean fraction of sites satisfying $|\phi(\mathbf{x},0)| > \phi_b$. (b) Mean largest cluster size of the barrier-crossing region. Error bars indicate standard errors over stochastic realizations. Both quantities increase monotonically with the initial-state label.

> Larger initial-state labels contain more and larger barrier-crossing seeds. However, those seeds are less compact and more spatially extended.

This suggests a natural mechanism for why the nucleation probability need not follow the monotonic increase of the barrier-crossing seed amount. Noncompact seeds may have a larger interfacial cost relative to their area and may therefore be less efficient at developing into growing nuclei. At the same time, this interpretation should be read conservatively: compactness and R_g are simple geometric diagnostics, not direct measurements of a committor probability or of a critical-droplet free energy.

To display the separation between seed amount and seed geometry more directly, Fig. 5(a) shows the sample-level relation between the barrier-crossing seed fraction and the compactness of the largest seed cluster. The small points show individual realizations, while the large connected markers show label means with standard-error bars. This makes clear that the larger-label ensembles move toward larger seed amount but lower compactness, rather than simply producing uniformly better nucleation precursors. Quantitatively, the sample-level association between p_{seed} and C_{comp} is negative (Pearson $r = -0.504$, Spearman $\rho = -0.517$). Figure 5(b) shows the corresponding direct comparison between the

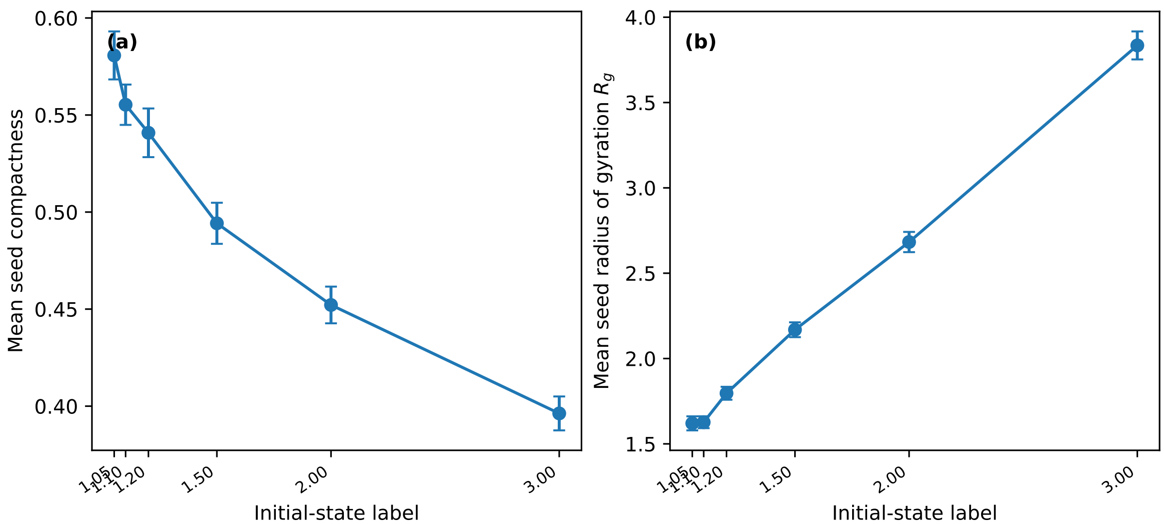

FIG. 4. Spatial quality of barrier-crossing seeds. (a) Mean compactness of the largest barrier-crossing seed cluster. (b) Mean radius of gyration of the largest barrier-crossing seed cluster. Higher initial-state labels generate larger but less compact and more spatially extended seeds.

detected cluster-nucleation time and the global transition time. For the 165 samples in which both times are observed, the correlation is very strong (Pearson $r = 0.997$, Spearman $\rho = 0.995$). The transition time is well described as a nucleation time followed by a growth delay, $t_{\mathrm{tr}} \simeq t_{\mathrm{nuc}} + \Delta t_{\mathrm{grow}}$, with $\langle \Delta t_{\mathrm{grow}} \rangle = 43.53 \pm 4.28$.

E. Robustness checks

We performed several robustness checks to verify that the above conclusions are not artifacts of a particular numerical or operational choice. First, we examined ordered-like seeds defined by $|\phi(\mathbf{x}, 0)| > 0.5\phi_s$. As shown in Fig. S1 of the Supplemental Material, the ordered-like seed fraction and the largest ordered-like cluster size are almost zero. This indicates that the observed nucleation events are unlikely to be deterministic growth of pre-existing ordered-phase droplets. Instead, nucleation is more naturally interpreted as a stochastic event formed during post-quench dynamics.

Second, we varied the cluster threshold used to identify nucleation events. As shown in Fig. S2(a), changing the threshold modifies the absolute nucleation probability, but the qualitative conclusion remains unchanged: the observed nucleation probability is not captured by seed amount alone.

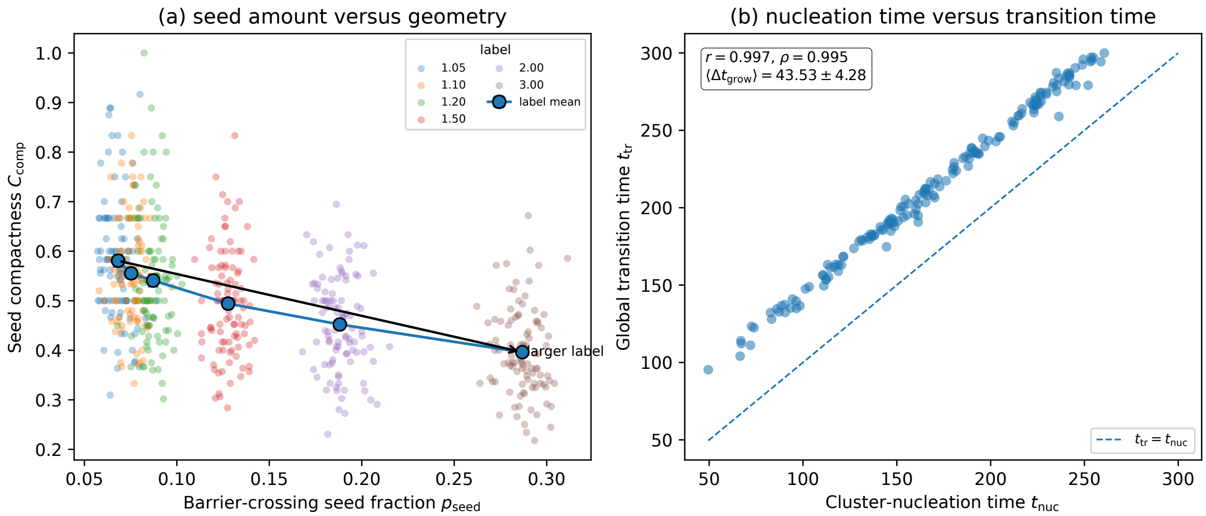

FIG. 5. Sample-level seed geometry and transition timing. (a) Relation between the barrier-crossing seed fraction p_{seed} and compactness C_{comp} of the largest barrier-crossing seed cluster. Small points are individual realizations, while large connected markers show label means with standard-error bars and indicate the trend with increasing initial-state label. (b) Direct comparison between the cluster-nucleation time t_{nuc} and the global transition time t_{tr} for samples in which both are observed. The dashed line indicates $t_{\mathrm{tr}} = t_{\mathrm{nuc}}$; the annotation gives the Pearson and Spearman correlations and the mean growth delay.

Third, we reduced the time step from $\Delta t = 0.02$ to $\Delta t = 0.01$. As shown in Fig. S2(b), the nucleation probability remains of the same order, and the qualitative seed statistics are preserved.

Finally, we checked the system size and pre-equilibration length. As shown in Fig. S3, the increase of p_{seed} and the decrease of seed compactness with the initial-state label are preserved in an $N = 128$ system-size check and in simulations with longer pre-equilibration. These robustness checks support the central conclusion that larger initial-state labels generate more barrier-crossing seeds, but those seeds are less compact and more spatially extended. Bootstrap and permutation trend checks reported in the Supplemental Material further confirm that these label-level trends are robust in the present data set. A lattice perimeter-to-area diagnostic provides an additional geometric measure of the largest barrier-crossing cluster.

As an additional sample-level check, we fitted logistic models for whether a cluster nucleation event was observed by the final observation time using seed-amount variables, seed-geometry variables, and their combination. The details are reported in the Supplemental Material. The seed-geometry-only model has the lowest Akaike information criterion in this check, but its improvement over an intercept-only model is modest. In the combined model, seed compactness has a positive standardized coefficient close to conventional significance, whereas p_{seed}, c_{seed}, and R_g do not provide statistically decisive predictors. We therefore do not interpret the logistic analysis as a committor calculation. Its role is instead to support the more conservative conclusion that seed amount and seed geometry should not be conflated. A finite-time restart analysis reported in the Supplemental Material reaches a similar conclusion: the committor-like nucleation probability varies across frozen initial configurations, but the present simple seed-geometry observables show only weak-to-modest correlations with it.

IV. DISCUSSION

The main contributions of this work are threefold. First, it separates continuous-transition and first-order transition protocols within the same spatial TDGL framework, showing that the relevant transition-time physics changes from deterministic growth of unstable modes to stochastic nucleation out of a metastable state. Second, it distinguishes barrier-crossing seed amount from seed geometry and shows that these quantities change in opposite directions as the initial-state label is increased. Third, it combines robustness checks, perimeter-based geometry diagnostics, finite-time binomial uncertainty estimates, and finite-time restart tests to delimit the interpretation of seed geometry: compactness and spatial extent are robust morphology diagnostics, but they are not by themselves complete committor-level predictors.

The present data do not establish compactness as a sufficient predictor of nucleation. Rather, they show that any reduced description based only on seed amount is incomplete, because seed amount and seed geometry vary independently across the initial ensembles.

A. Main interpretation: barrier-crossing seeds are not nuclei

The central message of the present work is that a barrier-crossing seed candidate should not be identified with a growing nucleus. A site or cluster satisfying $|\phi| > \phi_b$ has crossed the local free-energy barrier in field amplitude, but nucleation in a spatially extended first-order transition is not determined by local amplitude alone. A new-phase region must also overcome the interfacial free-energy cost associated with forming a boundary with the surrounding metastable phase [12].

This distinction explains the otherwise counterintuitive combination of results. The fraction of barrier-crossing sites $p_{\rm seed}$ and the largest barrier-crossing cluster size $c_{\rm seed}$ both increase monotonically with the initial-state label. If local barrier crossing were sufficient for nucleation, one would expect the nucleation probability to exhibit the same clear monotonic trend. Instead, the nucleation probability does not show such a trend within the present statistics. A natural candidate for the missing information is the spatial quality of the seed candidate.

In the present simulations, increasing the initial-state label produces more and larger barrier-crossing clusters, but these clusters become less compact and more spatially extended. Such clusters are not necessarily efficient nuclei. A fragmented or elongated cluster can have a large interfacial cost relative to its area, making it more susceptible to shrinking or being washed out by subsequent dynamics. Thus, seed amount and seed quality are distinct observables and should be analyzed separately when interpreting nucleation-controlled transition times.

B. Competition between seed amount and seed quality

The results suggest a competition between two effects. On one hand, larger initial-state labels enhance the probability of large field fluctuations and therefore increase the amount of barrier-crossing seed. On the other hand, the same increase in fluctuation strength generates seed clusters that are geometrically less compact. The initial state therefore changes both the number of nucleation candidates and their ability to act as growing nuclei.

This competition provides a natural mechanism by which a first-order transition can fail to show a simple monotonic relation between the initial-state label and the transition time.

A state with fewer but more compact seed candidates may compete with a state containing many extended, irregular barrier-crossing regions. In this sense, the relevant variable is not simply how much of the initial field has crossed the local barrier, but how much of that barrier-crossing field is organized into spatial structures that can survive the interfacial penalty.

The compactness used here is a deliberately simple geometric proxy rather than a direct measurement of interfacial free energy. Nevertheless, the monotonic decrease of compactness and the monotonic increase of the radius of gyration provide consistent evidence that the geometry of the largest barrier-crossing cluster changes systematically with the initial-state label. A sample-level logistic check reported in the Supplemental Material gives only modest predictive support for compactness, so the strongest conclusion is not that compactness alone determines nucleation. Rather, the robust conclusion is that seed geometry changes systematically and cannot be reduced to seed amount alone.

C. Role of post-quench stochastic dynamics

The ordered-like seed analysis further constrains the mechanism. Ordered-like seeds, defined by $|\phi| > 0.5\phi_s$, are almost absent in the initial state. Therefore, the observed nucleation events are unlikely to be deterministic growth of pre-existing stable-phase droplets. Instead, the initial field should be regarded as a spatially structured fluctuation landscape in which post-quench stochastic dynamics subsequently produces persistent ordered clusters.

This interpretation is consistent with the broad distribution of nucleation times and with the survival curves in Fig. 2(b). It is also supported by the direct comparison of t_{nuc} and t_{tr} in Fig. 5(b). The data show that, once both times are observed, the global transition time closely follows the stochastic nucleation time with a subsequent growth delay. The initial-state label controls the statistical structure of the field at the moment of the quench, while the post-quench stochastic dynamics controls whether and when a persistent nucleus appears.

A useful way to state the mechanism is therefore the following. The initial state does not usually supply an already-formed ordered nucleus. Rather, it supplies barrier-crossing fluctuations with a label-dependent spatial structure. Nucleation then occurs through post-quench stochastic dynamics, and the probability of nucleation depends on whether those fluctuations are spatially organized in a way that can develop into a growing cluster.

D. Relation to phase-transition Mpemba effects

The present study is not intended as a direct model of water freezing. Water involves evaporation, convection, dissolved gases, impurities, supercooling, hydrodynamic transport, latent heat, and microscopic crystallization processes [2–4]. The purpose of the present TDGL model is more limited: it isolates the role of nucleation-controlled transition times in a spatially extended order-parameter model.

Within this limited scope, the results are relevant to phase-transition Mpemba-like behavior. In continuous transitions and many Markovian relaxation problems, anomalous relaxation can often be discussed in terms of overlaps with slow modes or relaxation pathways [6, 7, 9]. In contrast, first-order transitions introduce metastability and nucleation. The transition time may then depend on rare spatial structures rather than only on global relaxation modes. This makes seed statistics a natural additional ingredient.

The present work also complements studies emphasizing spatial correlations in spin systems [19, 20]. Our results point to a related but more specific aspect of spatial structure: not merely whether correlations exist, but whether the initial field contains barrier-crossing regions with geometry favorable for nucleation. This seed-quality viewpoint may be useful for distinguishing different mechanisms that can appear in phase-transition protocols motivated by the Mpemba effect.

E. Anticipated objections and robustness of the conclusion

A first possible objection is that the conclusion might depend on the operational definition of nucleation. To address this, we varied the cluster threshold used to identify nucleation events. The absolute nucleation probability changes with this threshold, as expected, but the qualitative conclusion remains unchanged: nucleation is not predicted by the amount of barrier-crossing seed alone.

A second possible objection is that the compactness trend might be a numerical artifact of the discretization or of insufficient initial equilibration. We therefore checked a smaller time step, a larger system size, and a longer pre-equilibration time. These tests preserve the main trends: p_{seed} and c_{seed} increase with the initial-state label, while seed compactness decreases. These checks do not prove full finite-size convergence, but they make it unlikely that the seed-quality trend is an artifact of one numerical setting.

A third possible objection is that the present statistics are not sufficient to identify a unique optimal initial-state label. We agree. The detailed label dependence of the nucleation probability remains noisy, and we therefore do not claim that any particular initial-state label is optimal. The robust conclusion is more conservative: the observed finite-time nucleation probability is not captured by the monotonic increase in barrier-crossing seed amount.

A fourth possible objection is that a sample-level predictive analysis gives only limited evidence that compactness alone predicts nucleation. This is also correct. In the Supplemental Material, a logistic model using seed compactness and R_g gives the lowest AIC among the tested low-dimensional models, but its improvement over the intercept-only model is small. In a combined model, the compactness coefficient is positive and close to conventional significance, while the other seed variables are not statistically decisive. We therefore do not claim a quantitatively predictive nucleation-rate model from these observables alone.

A fifth possible objection is that compactness is not the same as the true committor probability or the exact critical droplet free energy. This is also correct. Compactness is used here as a simple and transparent geometric measure. A more quantitative nucleation analysis would require committor calculations, direct estimates of critical droplet shapes, or free-energy-based cluster analysis [24–26]. The present result should therefore be read as evidence that seed geometry is a necessary diagnostic to track, not as a complete theory of nucleation rates.

F. Limitations

Several limitations should be kept in mind. First, the initial-state label used in this work is temperature-like but not a calibrated thermodynamic temperature. It controls the strength of initial fluctuations in a minimal model. Quantitative comparison with an experimental cooling protocol is therefore not intended.

Second, the model is a two-dimensional scalar TDGL model with a nonconserved order parameter. It does not include hydrodynamics, latent heat transport, anisotropic crystal growth, conserved density fields, or microscopic molecular structure. The results should be regarded as a minimal theoretical study of nucleation-controlled transition times rather than a direct simulation of freezing.

Third, the definitions of nucleation time, ordered-like seed, and compactness are operational. We have checked robustness against several changes, but other definitions may emphasize different aspects of the dynamics. In particular, future work should compare the present geometric seed measures with committor-based or free-energy-based measures of nucleation probability.

Fourth, most of the main statistics are obtained at finite system size and over a finite number of stochastic samples. The robust trends in seed amount and seed geometry are clear, but precise estimates of nucleation probabilities, rare-event tails, and sample-level predictive relations would require larger ensembles.

G. Outlook

The present work suggests several directions for future study. One natural direction is to quantify the initial spatial structure more directly, for example through structure factors, correlation lengths, or cluster-shape distributions. Another is to vary the parameters a, b, and c in the ϕ^6 potential in order to tune the strength of the first-order transition and the nucleation barrier.

It would also be useful to replace the present compactness proxy with more direct measures of nucleation quality, such as perimeter-to-area ratio, critical cluster analysis, or committor probabilities. Finally, extending the analysis to models with conserved fields, hydrodynamic couplings, or explicit latent-heat transport may help connect the present minimal picture to more realistic phase-change systems.

V. CONCLUSION

We investigated transition times in spatially extended time-dependent Ginzburg–Landau models motivated by phase-transition Mpemba effects, focusing on the contrast between continuous and first-order transitions. The continuous-transition ϕ^4 model was used as a baseline. Under the present initialization, its transition time increases with the initial-state label, and a simple Mpemba-like inversion is not observed.

In the first-order ϕ^6 model, the post-quench state is metastable and the transition proceeds through stochastic nucleation. Larger initial-state labels generate more barrier-crossing seeds, as measured by both the seed fraction p_{seed} and the largest seed cluster size c_{seed}. However, the observed finite-time nucleation probability does not follow this monotonic increase.

The key observation is that seed amount and seed quality change in different directions. As the initial-state label increases, barrier-crossing seeds become more numerous and larger, but also less compact and more spatially extended. Ordered-like seeds close to the stable phase are almost absent in the initial state. These results indicate that nucleation is not simply deterministic growth of pre-existing ordered droplets, but a stochastic post-quench process influenced by the spatial structure of the initial field.

A direct comparison between nucleation and transition times further supports this picture: when both t_{nuc} and t_{tr} are observed, the global transition time is strongly correlated with the nucleation time. Thus, in the first-order model, transition times are tied to stochastic nucleation and subsequent growth rather than to seed amount alone.

The main conclusion of this work is that nucleation-controlled transition dynamics in phase-transition settings motivated by Mpemba-like relaxation cannot be characterized by seed amount alone. Seed quality, especially compactness and spatial extent, is an important diagnostic of whether barrier-crossing fluctuations are organized in a way that may develop into growing nuclei. A fully quantitative prediction of nucleation probabilities will require larger ensembles and more direct committor- or free-energy-based analyses.

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DATA AVAILABILITY

The simulation data, figure-generation scripts, and reproducibility instructions are available at `https://github.com/ssuzuki1017/tdgl-mpemba-seed-quality`. An archival

DOI will be added after the final repository release.

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